

Lukas Herajt

AI Engineer · RAG · Agents · MCP | Sioux Falls, SD | lukasherajt@gmail.com

SUMMARY

AI engineer with 5+ years shipping production software and 3+ years building LLM systems in production. Specialize in retrieval-grounded agent platforms, eval harnesses, and the routing/fallback policies that separate demos from products. Recent focus: Anthropic's Model Context Protocol (MCP), hybrid retrieval, and structured-output reliability for multi-step agent workflows.

EXPERIENCE

AI Engineer (Contract) — Title

Mid-2025 – Present · Remote

- Architecting production-grade LLM platforms with retrieval grounding, hybrid search, and per-tenant token budgets.
- Built MCP servers exposing internal APIs, vector stores, and knowledge bases to Claude agents with audit-logged tool calls.
- Designed latency- and cost-aware router across Claude Sonnet/Opus/Haiku and GPT-4 with deterministic fallbacks and circuit breakers.

AI Software Engineer — American Board of Anesthesiology

2022 – 2025

- Engineered token-cost optimization layers and evaluation harnesses for multi-step agentic workflows.
- Built LangChain agent logic on Anthropic and OpenAI function calling for extraction, classification, and structured JSON output.
- Shipped end-to-end document retrieval using OpenSearch, Pinecone, and FAISS with metadata filtering and hybrid retrieval — measurably reducing hallucinations across internal tools.
- Stood up accuracy checks, sampling tests, trace logs, and cost dashboards to monitor model stability and operational overhead.

Software Engineer — Mid Atlantic Distribution

2020 – 2022

- Built Python backend modules for data transformation, analytics, and ML-assisted automation.
- Delivered REST endpoints powering model-driven logic in business-facing tools.

Software Engineer — Hiretalk

2020

- Full-stack work across frontend and backend; first production role.

SELECTED PROJECTS

- **RAG Search Platform** — Embedding pipelines, OpenSearch + Pinecone + FAISS, hybrid retrieval with metadata filtering.
- **Agentic Document Workflows** — LangChain agents with Anthropic/OpenAI function calling, validation, and confidence checks.
- **Eval Kit** — Accuracy checks, sampling tests, cost dashboards, structured prompts with retry/fallback routing.
- **Model Router** — Latency- and cost-aware routing across Claude and GPT-4 with circuit breakers and per-tenant budgets.
- **MCP Tools** — Model Context Protocol servers exposing internal APIs and knowledge bases to Claude agents.

STACK

LLM & Agents: Anthropic Claude, OpenAI, MCP, LangChain, function calling, structured output, evals, prompt design

Machine Learning: Embeddings, vector search (Pinecone, FAISS, OpenSearch), retrieval grounding, hybrid search

Backend & Infra: Python, FastAPI, REST, PostgreSQL, Docker, observability, cost dashboards, CI/CD